

LIFI THE WIRELESS COMMUNICATION

A PROJECT REPORT

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## ABSTRACT

In this paper, a design of an intelligent prototype module for Battery management system of Electric vehicle. World is moving with automobiles. A huge amount of fossil fuels are burned for automobiles. The increase in air pollution and vehicular emissions has led to development of the renewable energy based generation and electrification of transportation. Recently, E-vehicle technology led to the booming of the automobile industry. EV plays an important role in greenhouse limitation and carbon pollution. In addition renewable energy sources like PV based onboard charging of EV is highly preferable for remote locations to improve lifespan of battery and efficient usage of onboard charging automated dual battery storage is proposed. To improve the Quality of battery and safety, it plays a vital role in the application of Electric mobility.

This Project reviews the attributes of the battery management system and its technology with advantages and disadvantages for EV applications. This project includes the battery voltage sensor, Microcontroller and solar panel charging to improve the performance of the battery for EV applications.

Batteries with better lifespan with the help of continuous monitoring of soc and doc of the batteries. If the two batteries of the dual battery system gets drained or fault intimation on the fault and location of the vehicle will be using the IoCTo mntordoulller. to ensure the efficiency of the proposed system. By using a

voltage sensor we can monitor the voltage level of the batteries. This EV contains

two batteries. When the main battery drains, this system will use the backup battery

through relay.Nothing on the earth is free of cost , but what if we could find a way

to implement free rides? A solar power<sub>v</sub> is a packaged, connected assembly of solar

cells, also called photovoltaic cells, which are solid state devices that can convert solar energy directly into electrical energy. Solar power generation plays a vital role in industrial as well as domestic applications. It is a cheap, Pollution free and green energy resource. Solar panel is used as a source for a backup battery. If two batteries get drain or fault, this intimation and location will update to webpage via IoT module, then the alert message will be sent to family members. This system status will display on the Liquid crystal display

## TABLE OF CONTENTS

|   |                                      |      |
|---|--------------------------------------|------|
|   | ACKNOWLEDGEMENT                      | iii  |
|   | ABSTRACT                             | iv   |
|   | LIST OF FIGURES                      | viii |
|   | LIST OF TERMS AND ABBREVIATIONS      | ix   |
| 1 | INTRODUCTION                         | 1    |
|   | 1.1 INTRODUCTION                     | 1    |
|   | 1.2 MOTIVATION FOR THE PROPOSED WORK | 4    |
|   | OBJECTIVES OF THE PROJECT            | 5    |
|   | ORGANIZATION OF THE PROJECT          | 5    |
|   | 1.3 1.4                              |      |
| 2 | LITERATURE SURVEY                    | 6    |
| 3 | EXISTING SYSTEM                      | 11   |
| 4 | PROPOSED SYSTEM                      |      |
|   | 4.1 PROPOSED SYSTEM                  | 12   |
|   | 4.2 BLOCK DIAGRAM                    | 13   |
|   | HARDWARE REQUIREMENTS                |      |
| 5 | 5.1                                  | 14   |
|   | 4*4 KEYPAD                           | 18   |
|   | 5.2 LCD MODULE 5.3                   | 20   |
|   | LED WITH 2 200HM 5.4                 | 22   |
|   | 5.5 AURDINO UNO                      | 23   |
|   | AUR DINO                             | 26   |
|   | NANO(RECEIVER)                       |      |

|   |                                            |    |
|---|--------------------------------------------|----|
| 6 | SOFTWARE REQUIREMENTS                      |    |
|   | 6.1 SOFTWARE DESCRIPTION                   |    |
|   | 6.1.1 ARDUINO IDE                          | 34 |
|   |                                            | 47 |
|   |                                            | 55 |
| 7 | RESULTS AND DISCUSSION                     |    |
|   | 7.1 EXPERIMENTAL INVESTIGATION AND RESULTS | 60 |
|   | 7.2 PROGRAM                                | 60 |
|   | 7.3 SOFTWARE RESULT                        | 67 |
| 8 | CONCLUSION                                 | 69 |
|   | REFERENCES                                 | 70 |
| 9 |                                            |    |

## LIST OF TERMS AND ABBREVIATIONS

|        |                                                |
|--------|------------------------------------------------|
| EV     | Electric Vehicle                               |
| BMS    | Battery Management System                      |
| PV     | Photo Voltaic                                  |
| MPP    | Maximum Power Point                            |
| RES-EV | Renewable Energy Source                        |
| IOT    | Internet of Things                             |
| SOC    | State Of Charge                                |
| SOH    | State Of Health                                |
| MPC    | Multiple Power Compensator                     |
| EPT    | Electric Power Train                           |
| ISO    | International Organization For Standardization |
| ADC    | Analog-To-Digital                              |
| DAC    | Digital-To-Analog                              |
| RISC   | Reduced Instruction Set Computer               |
| RTC    | Real Time Clock                                |
| SRAM   | Static Random Access Memory                    |
| MAC    | Media Access Control                           |
| LAN    | Local Area Network                             |
| SDIO   | Secure Digital Input Output                    |
| DC     | Direct Current                                 |

|          |                                                     |
|----------|-----------------------------------------------------|
| LCD SDK  | Liquid Crystal Display                              |
| AGC      | Software Development Kit                            |
| DDR      | Automatic Gain Control                              |
| LVDS     | Double Data Rate                                    |
| AVR      | Low Voltage Differential Signalling                 |
| EEPRO    | Automatic Voltage Regulator                         |
| M    DSP | Electrically Erasable Programmable Read Only Memory |
| MAC      | Digital Signal Processing                           |
|          | Multiply Accumulate Point                           |

# INTRODUCTION

## 1.1 INTRODUCTION

Li-Fi, or Light Fidelity, is a wireless communication technology that uses visible light, ultraviolet, or infrared spectrum to transmit data. Unlike Wi-Fi, which relies on radio waves, Li-Fi modulates light signals from LED bulbs to send information at high speeds, offering potential advantages like greater bandwidth, lower latency, and enhanced security due to light's inability to penetrate walls. It was first introduced by Professor Harald Haas in 2011 during a TED Talk, where he demonstrated data transmission through a flickering light bulb. Li-Fi is being explored for applications in environments where radio frequencies are limited, such as hospitals, aircraft, or dense urban areas, but it requires line-of-sight and

is still in developmental stages for widespread adoption.

Li-Fi, short for Light Fidelity, is an innovative wireless communication technology that utilizes the visible light spectrum, ultraviolet, or infrared light to transmit data, offering a high-speed alternative to traditional radio-frequency-based systems like Wi-Fi. Conceived by Professor Harald Haas at the University of Edinburgh and first showcased in his 2011 TED Global Talk, Li-Fi operates by modulating the intensity of light emitted from LED bulbs at imperceptible speeds to encode and transmit data. A photodetector, such as a smartphone camera or specialized receiver, captures these light signals and decodes them into usable information.

**High-Speed Data Transmission:** Li-Fi leverages the vast, unregulated visible light spectrum, which is 10,000 times larger than the radio spectrum, enabling theoretical speeds up to 100 Gbps in lab conditions—significantly faster than typical Wi-Fi networks.

**Mechanism:** Data is transmitted by rapidly switching LEDs on and off (flickering) at nanosecond intervals, invisible to the human eye. This modulation encodes binary data, which is then received and interpreted by light-sensitive devices.

**Line-of-Sight Requirement:** Since light cannot penetrate walls, Li-Fi requires a direct or reflected line of sight between the transmitter (LED) and receiver, which limits its range but enhances security by confining signals to a specific area.

**Dual Functionality:** Li-Fi-enabled LED lights can simultaneously provide illumination and data connectivity, making it energy-efficient and practical for smart lighting systems.

Since its introduction, Li-Fi has progressed from a novel concept to practical implementations. Companies like pureLiFi, Signify (Philips Lighting), and Oledcomm are developing Li-Fi-enabled products, including dongles, access points, and integrated lighting systems. Pilot projects have been deployed in schools, offices, and public spaces, with trials in Dubai, France, and the UK showcasing real-world viability. For instance, pureLiFi's systems have achieved speeds of 43 Mbps in commercial settings, and Signify's Trulifi system is being used in smart buildings.

However, Li-Fi is not poised to replace Wi-Fi entirely but rather to complement it in hybrid networks. Its adoption hinges on standardization, cost reduction, and device compatibility. The IEEE 802.11bb standard, released in 2023, has formalized Li-Fi protocols, paving the way for broader integration. Research is also advancing in using infrared or ultraviolet for non-visible light communication, mitigating issues like ambient light interference.

**Limited Range and Coverage:** The line-of-sight requirement and inability of light to penetrate obstacles restrict Li-Fi's coverage compared to Wi-Fi, necessitating multiple access points for seamless connectivity.

**Infrastructure Costs:** Widespread adoption requires retrofitting existing lighting systems with Li-Fi-compatible LEDs and installing receivers in devices, which could be costly.

**Ambient Light Interference:** Bright sunlight or other light sources can interfere with Li-Fi signals, though advancements in filtering and infrared-based Li-Fi are addressing this.

**Bidirectional Communication:** While downlink speeds are impressive, uplink communication (from device to transmitter) often requires integration with other technologies like Wi-Fi or infrared, complicating implementation.

by MPP (Maximum power point) value of solar panels. It is a capacity of the solar panel and the higher value can make higher MPP[6].

This project includes the battery voltage monitoring, state estimation, fault analysis, data acquisition and solar panel charging to improve the performance of the battery for EV applications.

## 1.2 Motivation for the proposed project.

The motivation for pursuing a Li-Fi project stems from its potential to address critical challenges in wireless communication while unlocking new opportunities for innovation, efficiency, and sustainability. Below are key motivations driving the development and implementation of Li-Fi technology:

### 1. Addressing Spectrum Congestion

**Problem:** The radio frequency (RF) spectrum used by Wi-Fi, 4G, and 5G is increasingly congested, especially in urban areas, leading to slower speeds and unreliable connections.

**Li-Fi Solution:** By utilizing the vast, unregulated visible light spectrum (10,000 times larger than the RF spectrum), Li-Fi offers a virtually limitless bandwidth, enabling faster and more reliable data transmission. A Li-Fi project can help alleviate spectrum scarcity and meet the growing demand for high-speed internet.

### 2. Ultra-High-Speed Connectivity

**Need:** Emerging technologies like 8K streaming, augmented reality (AR), virtual reality (VR), and the Internet of Things (IoT) require ultra-fast, low-latency networks.

**Motivation:** Li-Fi has demonstrated theoretical speeds up to 100 Gbps in lab settings, far surpassing typical Wi-Fi capabilities. A Li-Fi project can push the boundaries of connectivity, enabling seamless, high-bandwidth applications in smart homes, offices, and entertainment systems.

### 3. Enhanced Security Concern: Wi-Fi signals can be intercepted through walls, posing security risks in

sensitive environments like banks, military bases, or corporate offices. Li-Fi Advantage: Since light cannot penetrate walls, Li-Fi signals are confined to a specific area, drastically reducing the risk of unauthorized access. A Li-Fi project can provide secure communication solutions for industries requiring robust data protection.

### 4. Interference-Free Communication

**Issue:** RF-based systems often face interference from other wireless devices, electromagnetic sources, or crowded networks, degrading performance.

**Motivation:** Li-Fi operates in the light spectrum, immune to RF interference, making it ideal for environments like hospitals, aircraft, or industrial facilities where electromagnetic interference is a concern. A Li-Fi project can deliver stable, uninterrupted connectivity in such settings.

### 5. Energy Efficiency and Sustainability Goal: Reducing energy consumption in communication systems aligns with global sustainability goals.

### 1.3 Objectives of the project

The main objectives of the project were:

The objectives of a Li-Fi project are tailored to harness the potential of Light Fidelity technology to address specific technical, commercial, or societal goals. These objectives can vary depending on the project's scope (e.g., research, commercial development, or social impact), but below is a comprehensive list of common objectives for a Li-Fi project:

#### 1. Develop High-Speed Data Transmission

Objective: Design and implement a Li-Fi system capable of achieving ultra-high-speed data transfer (e.g., multi-Gbps) to support bandwidth-intensive applications like 4K/8K streaming, AR/VR, or IoT ecosystems. Purpose: Demonstrate Li-Fi's superiority over Wi-Fi in terms of speed and bandwidth, enabling next-generation connectivity.

#### 2. Enhance Security in Wireless Communication

Objective: Create a Li-Fi network that leverages light's inability to penetrate walls to provide secure data transmission, minimizing the risk of interception or hacking. Purpose: Cater to industries requiring high security, such as banking, healthcare, or military applications.

## Chapter 2

### LITERATURE SURVEY

#### 1. EFFICIENT BATTERY MONITORING SYSTEM FOR E-VEHICLES

##### Abstract :

LIGHT FIDELITY (LI-FI) IS AN INNOVATIVE WIRELESS COMMUNICATION TECHNOLOGY THAT UTILIZES VISIBLE LIGHT FROM LIGHT-EMITTING DIODES (LEDs) TO TRANSMIT DATA, OFFERING A HIGH-SPEED, SECURE, AND ENERGY-EFFICIENT ALTERNATIVE TO TRADITIONAL RADIO FREQUENCY-BASED WI-FI. PROPOSED BY GERMAN PHYSICIST HARALD HAAS IN 2011, LI-FI OPERATES BY MODULATING THE INTENSITY OF LED LIGHT AT SPEEDS IMPERCEPTIBLE TO THE HUMAN EYE, ENABLING DATA TRANSFER RATES SIGNIFICANTLY FASTER THAN WI-FI, WITH POTENTIAL SPEEDS EXCEEDING 100 GBIT/S IN CONTROLLED ENVIRONMENTS. THIS PROJECT EXPLORES THE DESIGN AND IMPLEMENTATION OF A LI-FI SYSTEM, FOCUSING ON ITS ARCHITECTURE, WHICH INCLUDES LED-BASED TRANSMITTERS AND PHOTODIODE RECEIVERS, AND ITS APPLICATIONS IN ENVIRONMENTS WHERE RADIO INTERFERENCE IS A CONCERN, SUCH AS HOSPITALS, AIRPLANES, AND UNDERWATER COMMUNICATION. THE SYSTEM LEVERAGES THE VAST, UNLICENSED VISIBLE LIGHT SPECTRUM, PROVIDING HIGHER BANDWIDTH, ENHANCED SECURITY DUE TO LIGHT'S INABILITY TO PENETRATE WALLS, AND REDUCED ELECTROMAGNETIC INTERFERENCE. CHALLENGES SUCH AS LINE-OF-SIGHT REQUIREMENTS, SENSITIVITY TO AMBIENT LIGHT, AND UPLINK COMMUNICATION LIMITATIONS ARE ADDRESSED THROUGH HYBRID MODELS INTEGRATING WI-FI FOR UPLINK AND ADVANCED MODULATION TECHNIQUES. THE PROJECT DEMONSTRATES A PROTOTYPE FOR SHORT-RANGE, HIGH-SPEED DATA TRANSMISSION, HIGHLIGHTING LI-FI'S POTENTIAL TO ALLEVIATE SPECTRUM CONGESTION AND SUPPORT THE GROWING DEMAND FOR WIRELESS CONNECTIVITY IN THE ERA OF SMART CITIES AND THE INTERNET OF THINGS (IOT). FUTURE RESEARCH DIRECTIONS INCLUDE IMPROVING NON-LINE-OF-SIGHT PERFORMANCE AND DEVELOPING COST-EFFECTIVE MULTI-USER ACCESS SOLUTIONS.

Authors: M. Senthilkumar; K.P. Suresh; T.Guna Sekar; C. Pazhani Muthu

## Abstract:

Light Fidelity (Li-Fi) is a cutting-edge wireless communication technology that uses visible light spectrum (400–800 THz) to transmit data, offering a promising alternative or complement to traditional radio frequency (RF) technologies like Wi-Fi and cellular networks. Unlike Wi-Fi, which relies on electromagnetic waves in the 2.4 GHz or 5 GHz bands, Li-Fi modulates light intensity from LEDs to encode data, which is then detected by photodiodes or light sensors. The concept, first introduced by Harald Haas in his 2011 TED Talk, leverages the ubiquitous presence of LED lighting to create a dual-purpose system for illumination and high-speed data communication.

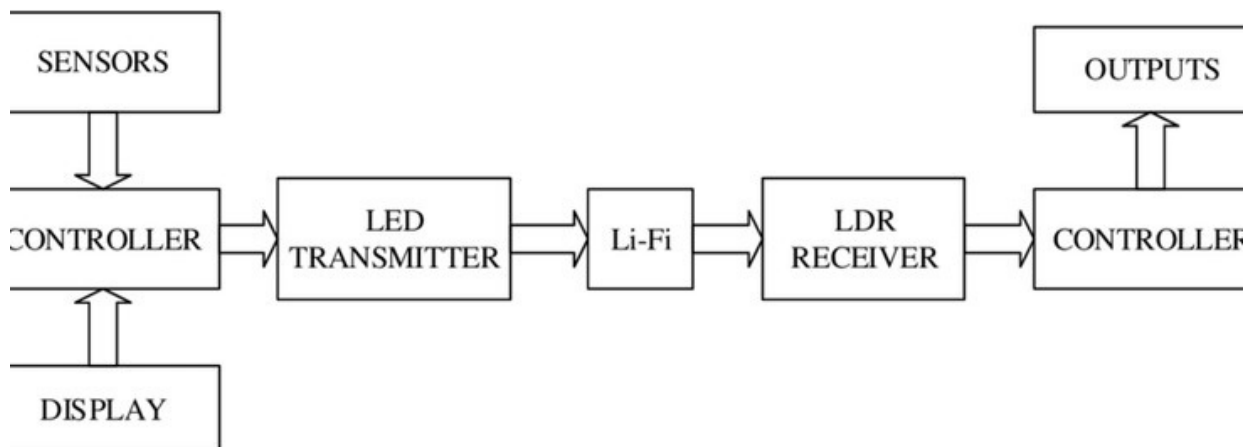
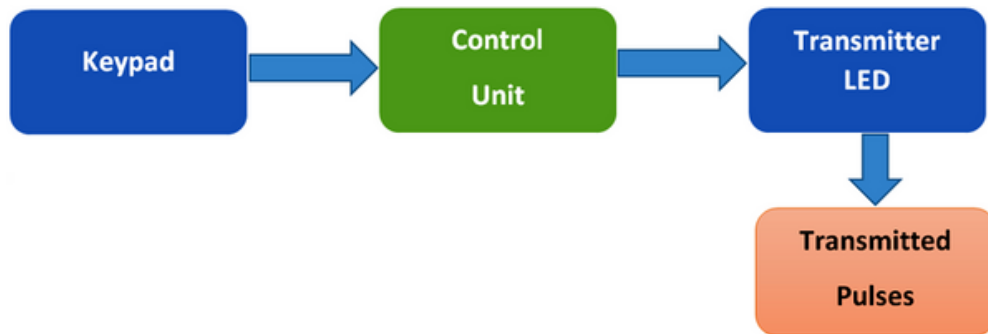
Li-Fi has gained attention due to the increasing congestion of the RF spectrum and the growing demand for secure, high-bandwidth connectivity driven by the Internet of Things (IoT), 5G, and smart infrastructure. With theoretical speeds exceeding 100 Gbit/s in lab settings (demonstrated by researchers at the University of Oxford in 2015), Li-Fi offers transformative potential for industries requiring reliable, interference-free communication.

The Li-Fi project underscores its transformative potential as the world moves toward 6G and beyond. Key areas for future development include:

- Non-Line-of-Sight Improvements: Using advanced optics, beamforming, or diffused light to extend coverage.
- Multi-User Access: Developing protocols for simultaneous connections, similar to Wi-Fi's MIMO (Multiple Input Multiple Output).
- Device Integration: Embedding Li-Fi receivers in smartphones, laptops, and IoT devices for widespread adoption.
- Hybrid Networks: Seamlessly integrating Li-Fi with Wi-Fi, 5G, or satellite systems for ubiquitous connectivity.
- Commercialization: Companies like pureLiFi and Signify are already deploying Li-Fi in niche markets, with broader adoption expected as costs decrease.

Author: **Harald Haas,**

## 4.2 BLOCK DIAGRAM



## Chapter 5

### HARDWARE REQUIREMENTS

#### 5.1 ARDUINO UNO

##### INTRODUCTION

Arduino is a great platform for beginners into the World of Microcontrollers and Embedded Systems. With a lot of cheap sensors and modules, you can make several projects either as a hobby or even commercial.

As technology advanced, new project ideas and implementations came into play and one particular concept is the Internet of Things or IoT. It is a connected platform, where several “things” or devices are connected over internet for exchange of information.

In DIY community, the IOT projects are mainly focused on Home Automation and Smart Home applications but commercial and industrial IoT projects have far complex implementations like Machine Learning, Artificial Intelligence, Wireless Sensor Networks etc.

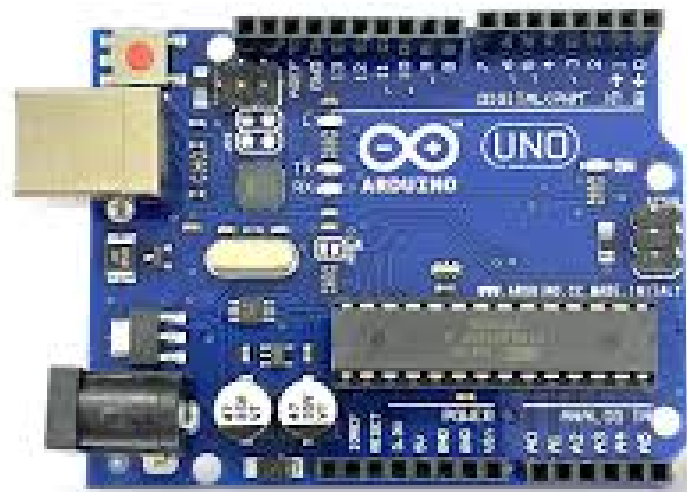
The important thing in this brief intro is whether it is a small DIY project by a hobbyist or a complex industrial project, any IoT project must have connectivity to Internet. This is where the likes of ESP8266 and ESP32 come into picture.

If you want to add Wi-Fi connectivity to your projects, then ESP8266 is a great option. But if you want build a complete system with Wi-Fi connectivity, Bluetooth

connectivity, high resolution ADCs, DAC, Serial Connectivity and many other features, then ESP32 is the ultimate choice.

The Arduino Uno is a popular open-source microcontroller board based on the ATmega328P chip, developed by Arduino.cc. It's widely used for beginners and hobbyists in electronics projects due to its simplicity and versatility. The board features 14 digital input/output pins (6 of which can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection for programming and power, a DC power jack, an ICSP header, and a reset button. It supports a variety of shields for added functionality, such as Bluetooth, Wi-Fi or motor control, and is programmed using the Arduino IDE with C/C++-based code. Operating at 5V, it can be powered via USB or an external 7-12V source. Its affordability, extensive community support, and compatibility with sensors and modules make it ideal for prototyping and learning.

Xtensa LX7 dual-core microprocessor or a single-core RISC-V microprocessor and includes built-in antenna switches, RF balun, power amplifier, low-noise receive amplifier, filters, and power-management modules. ESP32 is created and developed by Espressif Systems, a Shanghai-based Chinese company, and is manufactured by TSMC using their 40 nm process.



## Specifications of AURDINO

ESP32 has a lot more features than ESP8266 and it is difficult to include all the specifications in this Getting Started with ESP32 guide. So, I made a list of some of the important specifications of ESP32 here. But for complete set of specifications, I strongly suggest you to refer to the Datasheet.

- Single or Dual-Core 32-bit LX6 Microprocessor with clock frequency up to 240 MHz. 520 KB of SRAM, 448 KB of ROM and 16 KB of RTC SRAM.
- Supports 802.11 b/g/n Wi-Fi connectivity with speeds up to 150 Mbps.
- Support for both Classic Bluetooth v4.2 and BLE specifications.
- 34 Programmable GPIOs. Up to 18 channels of 12-bit SAR ADC and 2 channels of 8-bit DAC
- Serial Connectivity include 4 x SPI, 2 x I2C, 2 x I2S, 3 x UART. Ethernet MAC for physical LAN Communication (requires external PHY).
- 1 Host controller for SD/SDIO/MMC and 1 Slave controller for SDIO/SPI.
- Motor PWM and up to 16-channels of LED PWM.
- Secure Boot and Flash Encryption.
- Cryptographic Hardware Acceleration for AES, Hash (SHA-2), RSA, ECC and RNG.
- 
- 

## 5.2 ARDUINO NANO

### GENERAL DESCRIPTION

The Arduino Nano is a compact, breadboard-friendly microcontroller board based on the ATmega328P (or ATmega168 in older versions), developed by Arduino.cc. It's designed for projects where space is limited, offering similar functionality to the Arduino Uno in a smaller form factor. The Nano has 14 digital input/output pins (6 of which support PWM), 8 analog inputs, a 16 MHz crystal oscillator, a mini-USB port for programming and power, and an ICSP header. Measuring just 18 x 45 mm, it's ideal for portable or embedded applications. It operates at 5V and can be powered via USB or an external 6- 20V source (recommended 7-12V). Like other Arduino boards, it's programmed using the Arduino IDE with C/C++-based code and supports a wide range of sensors and modules. Its small size, low cost, and versatility make it a favorite for DIY electronics and prototyping.

## PRODUCT DESCRIPTION

The Arduino Nano is a compact, breadboard-friendly microcontroller board developed by Arduino.cc, based on the ATmega328P microcontroller (or ATmega168 in older versions). Released in 2008, it measures just 45 x 18 mm and weighs around 7 grams, making it ideal for projects where space is a constraint. Despite its small size, it offers functionality similar to the larger Arduino Uno, featuring 14 digital input/output pins (6 of which support PWM), 8 analog inputs, a 16 MHz crystal oscillator, a mini-USB port for programming and power, and an ICSP header for in-circuit programming. It operates at 5V, with an input voltage range of 7-12V (6-20V possible but not recommended beyond 12V), and each pin can handle a maximum current of 40mA.



aurdino nano

## FEATURES

- Under-voltage, over-voltage, or voltage band models

- Powered from sensing input lines or from separate AC supply
- Available with time delays on pull-in and/or drop-out or with customized voltage-time trip curves

## APPLICATIONS

- Power generation unit
- Power transmission unit
- Industry and other applications

## 5.3 LDR MODULE

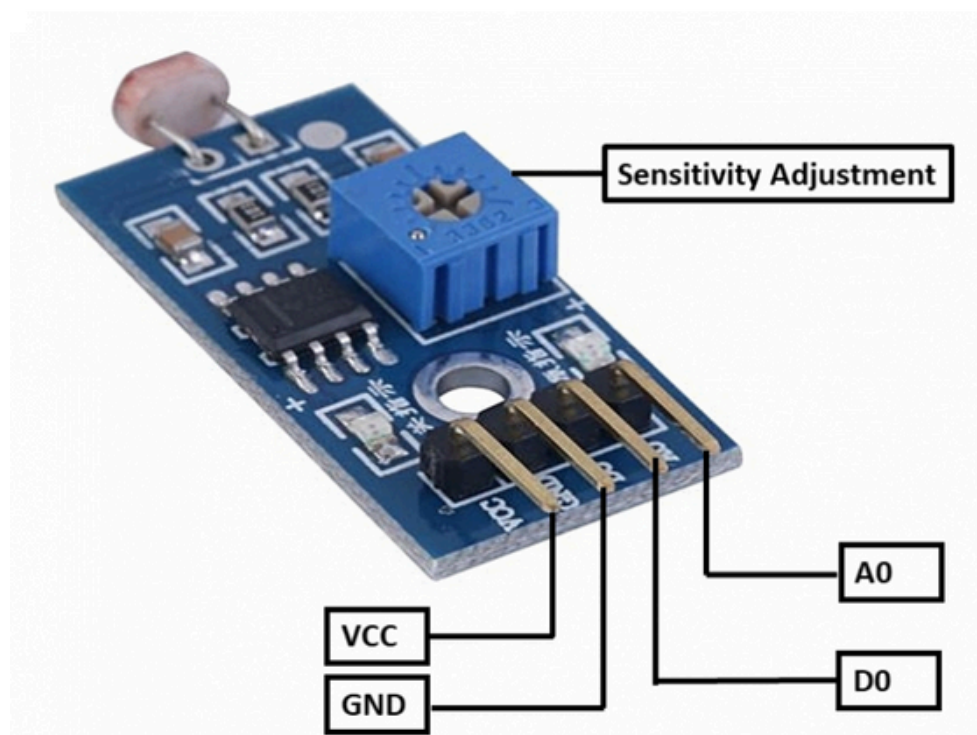
### GENERAL DESCRIPTION

An LDR (Light Dependent Resistor) module is a simple, cost-effective sensor used in electronics projects to detect light intensity. It typically consists of a photocell (the LDR itself) paired with supporting components like a resistor and sometimes a comparator (e.g., LM393) on a small PCB. The LDR, made of a semiconductor material like cadmium sulfide, changes its resistance based on the amount of light it receives—resistance decreases as light intensity increases (e.g., from several M $\Omega$  in darkness to a few hundred  $\Omega$  in bright light), and vice versa. The module usually has 3 or 4 pins: VCC (typically 3.3V or 5V), GND, and an output pin (DO for digital output, AO for analog output if available). The digital output switches between HIGH and LOW based on a threshold set by an onboard potentiometer, while the analog output provides a continuous voltage proportional to light intensity. Some modules include an LED indicator to show

when the threshold is crossed. Operating voltage is generally 3.3V to 5V, making it compatible with microcontrollers like Arduino Uno or Nano.

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LDR modules are widely used in light-sensitive applications, such as automatic streetlights, solar trackers, or, as in your case, LiFi systems for detecting light signals. They're inexpensive, easy to interface with (via digital or analog pins on a microcontroller), and require minimal setup—typically just a few lines of code to read values in the Arduino IDE. However, they're sensitive to temperature changes, have a slower response time (tens of milliseconds), and can be less accurate in extreme lighting conditions compared to advanced sensors like photodiodes. Their simplicity and versatility make them a popular choice for hobbyists and educational projects.



## 5.4 4\*4 keypad

### GENERAL DESCRIPTION

A 4x4 keypad is a compact input device commonly used in electronics projects for user interaction, featuring a grid of 16 push-button keys arranged in 4 rows and 4 columns. Each key typically represents a character or function, such as numbers (0-9), letters (A-D), or symbols (\*, #), making it versatile for applications like entering passwords, controlling devices, or navigating menus. The keys are mounted on a membrane or PCB, with a matrix configuration that reduces the number of microcontroller pins needed for interfacing—using 8 pins (4 for rows, 4 for columns) instead of 16. Internally, the keypad works by connecting rows and columns through a conductive path when a key is pressed, creating a unique electrical signal for each button. For example, pressing "5" connects row 2 and column 2. This matrix is scanned by a microcontroller (like Arduino Uno or Nano) to detect which key is pressed, often using libraries like "Keypad.h" in the Arduino IDE to simplify coding. The module operates at 5V or 3.3V, drawing minimal current, and connects directly to a microcontroller's digital pins.

Geared dc motors can be defined as an extension of dc motors. A geared DC Motor has a gear assembly attached to the motor. The speed of the motor is counted in terms of rotations of the shaft per minute and is termed as RPM .The gear assembly helps in increasing the torque and reducing the speed. Using the correct combination of gears in a gear motor, its speed can be reduced to any desirable figure. This concept where gears reduce the speed of the vehicle but increase its torque is known as gear reduction. A DC motor can be used at a voltage lower than the rated voltage. But, below 1000 rpm, the speed becomes unstable, and the motor will not run smoothly.



## FEATURES

A 4x4 keypad offers several key features that make it a popular choice for electronics projects:

- **16-Key Matrix Layout:** Arranged in 4 rows and 4 columns, providing 16 buttons labeled with numbers (0-9), letters (A-D), and symbols (\*, #) for versatile input options like numeric codes or menu navigation.
- **Compact Design:** Typically measures around 70 x 77 mm, lightweight (about 20-30 grams), and slim, making it easy to integrate into space-constrained projects.
- **Matrix Configuration:** Uses 8 pins (4 rows, 4 columns) for interfacing, reducing the number of microcontroller pins needed compared to individual button setups, optimizing I/O usage.
- **Low Power Consumption:** Operates at 3.3V to 5V, drawing minimal current (a few mA per key press), suitable for battery-powered applications.

### 5.4 **battery**

In electricity, a battery is a device consisting of one or more electrochemical cells that convert stored chemical energy into electrical energy. Since the invention of the first battery (or "voltaic pile") in 1800 by Alessandro Volta and especially since the technically improved Daniell cell in 1836, batteries have become a

common power source for many household and industrial applications. According to a 2005 estimate, the worldwide battery industry generates US\$48 billion in sales each year, with 6% annual growth.

There are two types of batteries: primary batteries (disposable batteries), which are designed to be used once and discarded, and secondary batteries (rechargeable batteries), which are designed to be recharged and used multiple times. Batteries come in many sizes from miniature cells used to power hearing aids and wristwatches to battery banks the size of rooms that provide standby power for telephone exchanges and computer data centers.

A battery is a device that converts chemical energy directly to electrical energy. It consists of a number of voltaic cells; each voltaic cell consists of two half-cells connected in series by a conductive electrolyte containing anions and cations. One half-cell includes electrolyte and the electrode to which anions (negatively charged ions) migrate, i.e., the anode or negative electrode; the other half-cell includes electrolyte and the electrode to which cations (positively charged ions) migrate, i.e., the cathode or positive electrode. In the redox reaction that powers the battery, cations are reduced (electrons are added) at the cathode, while anions are oxidized (electrons are removed) at the anode. The electrodes do not touch each other but are electrically connected by the electrolyte. Some cells use two half-cells with different electrolytes. A separator between half-cells allows ions to flow, but prevents mixing of the electrolytes.

Batteries are classified into two broad categories, each type with advantages and disadvantages.

● *Primary* batteries irreversibly (within limits of practicality) transform chemical energy to electrical energy. When the initial supply of reactants is exhausted, energy cannot be readily restored to the battery by electrical means.

● *Secondary* batteries can be recharged; that is, they can have their chemical reactions reversed by supplying electrical energy to the cell, restoring their original composition.

Some types of primary batteries used, for example, for telegraph circuits, were restored to operation by replacing the components of the battery consumed by the chemical reaction. Secondary batteries are not indefinitely rechargeable due to dissipation of the active materials, loss of electrolyte and internal corrosion.

Primary batteries

*Main article: primary cells*

Primary batteries can produce current immediately on assembly. Disposable batteries are intended to be used once and discarded. These are most commonly used in portable devices that have low current drain, are used only intermittently, or are used well away from an alternative power source, such as in alarm and communication circuits where other electric power is only intermittently available. Disposable primary cells cannot be reliably recharged, since the chemical reactions are not easily reversible and active materials may not return to their original forms. Battery manufacturers recommend against attempting recharging primary cells. Common types of disposable batteries include Zinc-carbon batteries and alkaline batteries. In general, these have higher energy densities than rechargeable batteries, but disposable batteries do not fare well under high-drain applications with loads under 75 ohms(75

$\Omega$ ). 25

## Secondary batteries

### *Main article: Rechargeable battery*

Secondary batteries must be charged before use; they are usually assembled with active materials in the discharged state. Rechargeable batteries or secondary cells can be recharged by applying electric current, which reverses the chemical reactions that occur during its use. Devices to supply the appropriate current are called chargers or rechargers.

The oldest form of rechargeable battery is the lead-acid battery . This battery is notable in that it contains a liquid in an unsealed container, requiring that the battery be kept upright and the area be well ventilated to ensure safe dispersal of the hydrogen gas produced by these batteries during overcharging. The lead–acid battery is also very heavy for the amount of electrical energy it can supply. Despite this, its low manufacturing cost and its high surge current levels make its use common where a large capacity (over approximately 10 Ah) is required or where the weight and ease of handling are not concerns.

## 5.6 16×2 LCD GENERAL DESCRIPTION

LCD stands for liquid crystal display. They come in many sizes 8x1 , 8x2 , 10x2 , 16x1 , 16x2 , 16x4 , 20x2 , 20x4 , 24x2 , 30x2 , 32x2 , 40x2 etc . Many multinational companies like Philips Hitachi Panasonic make their own special kind of LCD'S to be used in their products. All the LCD'S perform the same functions (display characters numbers, special ASCII characters etc).Their programming is also the same and they all have the same 14 pins (0-13) or 16 pins (0 to 15). Alphanumeric displays are used in a wide range of applications, including palmtop

computers, word processors, photocopiers, point of sale terminals, medical instruments, cellular phones, etc.

## PRODUCT DESCRIPTION

This is an LCD Display designed for E-blocks. It is a 16 character, 2-line alphanumeric LCD display connected to a single 9-way D-type connector. This allows the device to be connected to most E-Block I/O ports. The LCD display requires data in a serial format, which is detailed in the user guide below. The display also requires a 5V power supply. Please take care not to exceed 5V, as this will cause damage to the device. The 5V is best generated from the E-blocks Multi programmer or a 5V fixed regulated power supply. The 16 x 2 intelligent alphanumeric dot matrix displays is capable of displaying 224 different characters and symbols. A full list of the characters and symbols is printed on pages 7/8 (note these symbols can vary between brand of LCD used). This booklet provides all the technical specifications for connecting the unit, which requires a single power supply (+5V).



Fig.7 LCD

## FEATURES

- Input voltage: 5v

- E-blocks compatible
- Low cost
- Compatible with most I/O ports in the E-Block range
- Ease to develop programming code using Flow code icons

## APPLICATIONS

- Monitoring

### 2.6 LED

#### GENERAL DESCRIPTION

An LED (Light Emitting Diode) is a semiconductor device that emits light when an electric current passes through it, widely used in electronics for indicators, displays, and, in your case, LiFi systems for data transmission.

It consists of a p-n junction diode made from materials like gallium arsenide or gallium phosphide, which releases photons when electrons recombine with holes during current flow. The color of the light depends on the material's bandgap—red, green, blue, or white (using a phosphor coating) are common, with wavelengths ranging from 400 nm (violet) to 700 nm (red).

LEDs are typically small, around 5mm in diameter for standard types, with two leads: the anode (positive) and cathode (negative), the latter often marked by a flat edge or shorter lead. They operate at low voltages, usually 1.8V to 3.3V depending on the color (e.g., red LEDs at ~2V, blue at ~3V), and draw low current, around 20mA for standard LEDs, though high-power LEDs can handle more. A current-limiting resistor (e.g., 220Ω for a 5V supply) is often required to prevent burnout. They have a fast response time, switching on/off in nanoseconds, making them ideal for high-speed applications like LiFi.

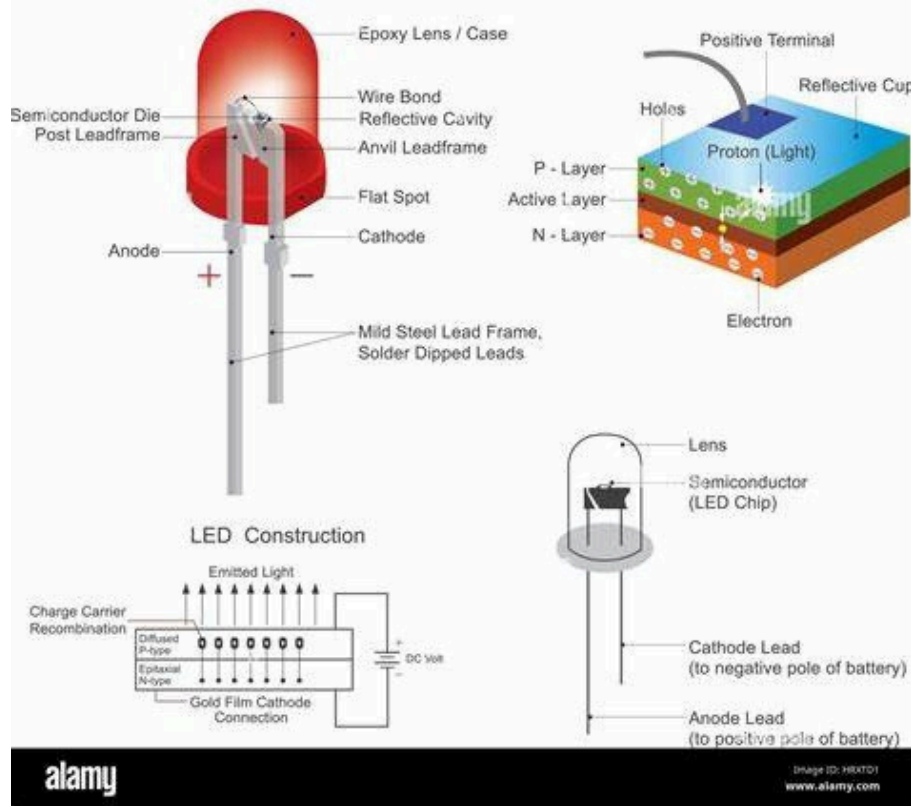
In projects, LEDs are used with microcontrollers like Arduino Uno or Nano, connected to digital pins for simple on/off control or PWM pins for brightness adjustment. They're highly efficient, converting most energy into light rather than heat, with a lifespan of 25,000 to 50,000 hours. LEDs are directional, emitting light in a narrow beam (typically 20-120 degrees), and are available in various forms—through-hole, SMD, or high-power modules. They're inexpensive, costing a few cents each, and are robust, resisting shock and vibration. However, they're polarity-sensitive, can be damaged by overcurrent, and may require heat sinks for high-power models. Their versatility, efficiency, and speed make them essential in lighting, displays, and communication systems.

Nowadays we have several small to even tiny sized solar panels and these solar panels are being used in such electrical devices as pocket calculators and to charge up cell phones. Small solar panels are portable and can be taken anywhere you want or need electrical power or battery recharging. You can see them in the car window sometimes as they are charging up the car battery. 12v solar panel cells are really semiconductors that are made using photovoltaic materials.

Photovoltaic (PV) is the technical terminology that is related to solar energy and the conversion of the sun's ultraviolet rays into electrical energy. They do not have fluids or chemicals in them and they do not have any moving parts. Solar panel cells simply convert the sunlight they receive into stored energy with the help of an inverter. You can take a portable solar panel with you when you travel to places that will not have electricity and use them.

A 12v solar panel can store energy from the battery so that it can be used when there is no sunlight available. You can use a 12v solar panel with various electrical appliances. They are a convenient power supply for charging mobile phones, personal stereos, PDAs and toys, etc. Some can be used for trickle

# A Light-Emitting Diode (LED)



## features

I'll assume "it" refers to the LED, as it was the last component described. Here are the key features of an LED (Light-Emitting Diode):

- **Efficient Light Emission:** Converts most electrical energy into light, not heat, with efficiency up to 80-90%, far better than incandescent bulbs (10-15%).
- **Low Power Consumption:** Operates at 1.8V to 3.3V and ~20mA for standard LEDs, making it ideal for battery-powered projects like LiFi systems.
- **Fast Response Time:** Switches on/off in nanoseconds (e.g., <10 ns), perfect for high-speed applications such as data transmission in LiFi.
- **Long Lifespan:** Lasts 25,000 to 50,000 hours, outliving traditional bulbs (1,000 hours for incandescent), reducing replacement needs.
- **Compact Size:** Standard LEDs are small (~5mm diameter), with SMD variants even smaller, fitting easily into tight spaces in circuits.

## ADVANTAGES OF LED :

Since the last component discussed was the LED, I'll provide the advantages of an LED in the context of your LiFi block diagram project using Arduino Uno, Nano, 4x4 keypad, LDR module, and LED.

- **High Efficiency:** LEDs convert most energy into light (80-90% efficiency), minimizing power usage, which is ideal for a LiFi system where the LED transmits data while keeping energy costs low for battery-powered Arduino setups. **Fast Switching Speed:** With a response time of nanoseconds ( $<10$  ns), LEDs can modulate light rapidly, enabling high-speed data transmission in LiFi, crucial for encoding and sending data via light signals to the LDR module. **Low Power Consumption:** Operating at  $\sim 20$ mA and 1.8V-3.3V, LEDs draw minimal power, extending the runtime of your Arduino Uno or Nano, especially in portable or low-power LiFi applications. **Long Lifespan:** Lasting 25,000 to 50,000 hours, LEDs ensure reliability in your LiFi system, reducing maintenance needs compared to traditional light sources that might fail sooner. **Compact and Lightweight:** At  $\sim 5$ mm in diameter, LEDs fit easily into your project's design, leaving space for other components like the 4x4 keypad and LDR module on a breadboard or PCB. **Minimal Heat Output:** LEDs produce little heat, preventing overheating issues in your setup, which is beneficial when working with sensitive components like the LDR module that might be affected by temperature changes. **Directional Light:** Emitting light in a focused beam (20-120 degrees), LEDs ensure efficient transmission to the LDR module in your LiFi system, reducing signal loss and improving data accuracy. **PWM Support:** LEDs can be dimmed or modulated using PWM pins
-

- As you can see, there are many advantages to solar energy. The advantages of solar energy range from benefiting your pocketbook to benefiting the environment. There are actually only a few features of solar energy that can be considered disadvantages.

## Chapter 6

### 6. SOFTWARE REQUIREMENTS

EMBEDDED C

ARDUINO IDE

IoT

### 6.1 SOFTWARE DESCRIPTIONS:

#### 6.1.1 Arduino Software (IDE)

Get the latest version from the download page. You can choose between the Installer (.exe) and the Zip packages. We suggest you use the first one that installs directly everything you need to use the Arduino Software (IDE), including the drivers. With the Zip package you need to install the drivers manually. The Zip file is also useful if you want to create a portable installation. When the download finishes, proceed with the installation and please allow the driver installation process when you get a warning from the operating system.

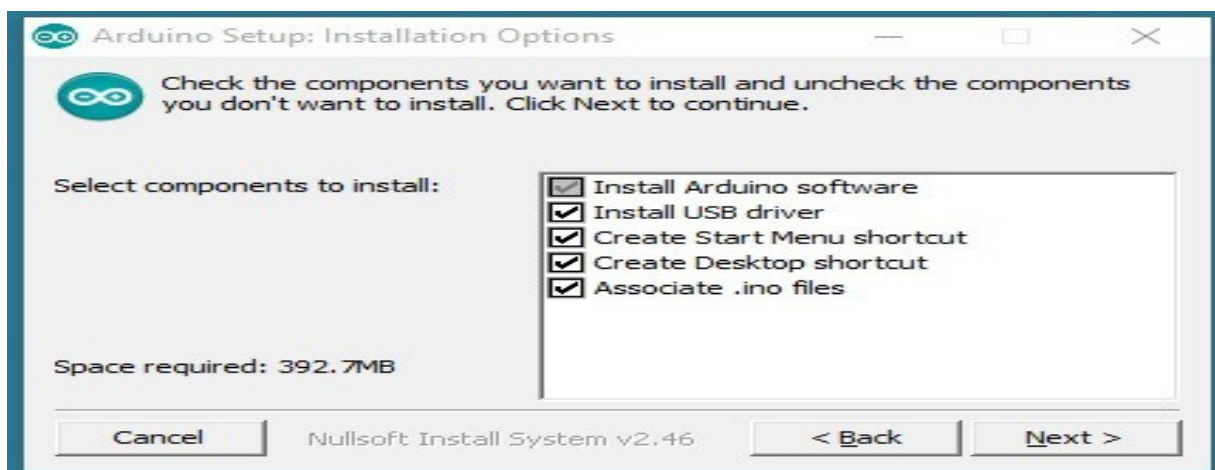


Fig .9 Choose the components to install

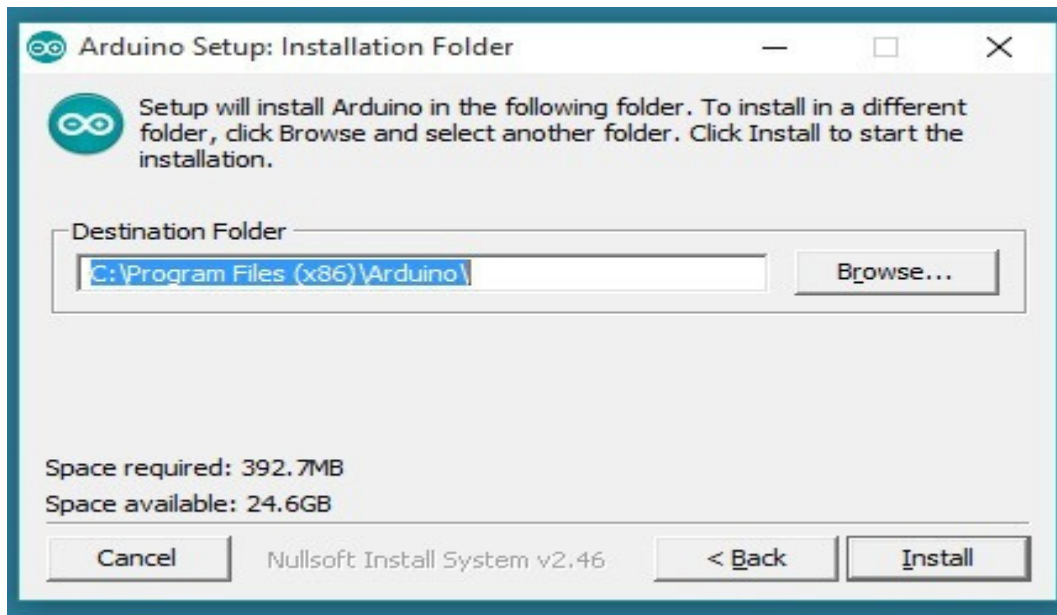


Fig 10 Choose the installation directory

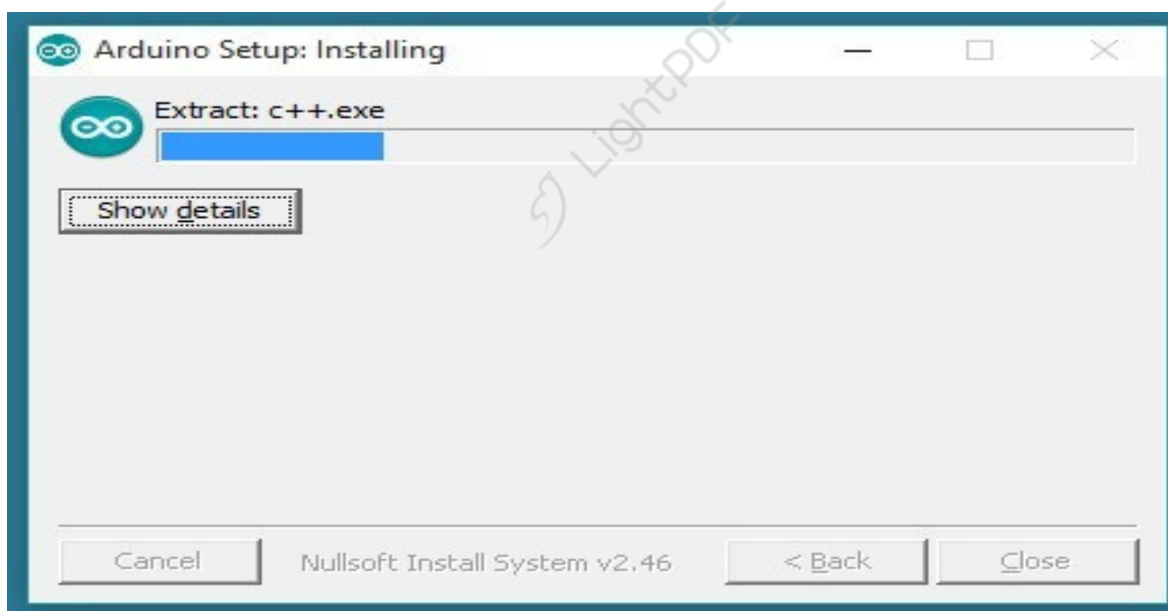


Fig.11ArduinoSoftware(IDE)

The process will extract and install all the required files to execute properly the Arduino Software (IDE)

Arduino Boot loader Issue

The current boot loader burned onto the Arduino UNO is not compatible with ROBOTC. In its current form, you will be able to download the ROBOTC Firmware to the ArduinoUNO, but you will not be able to download any user programs. The reason for this is because there is a bug in the Arduino UNO firmware that does not allow flash write commands to start at anywhere but the beginning of flash memory (0x000000). See the bottom of this page for more technical details.

Because ROBOTC is not able to burn a new bootloader as of today, you will need to use the Arduino's Open Source language with a modified bootloader file to re-burn your bootloader on your Arduino UNO boards. The enhanced bootloader is backwards compatible with the original one. That means you'll still be able to program it through the Arduino programming environment as before, in addition to ROBOTC for Arduino.

### Hardware Needed

To burn a new version of the Arduino boot loader to your UNO, you'll need an AVR ISP Compatible downloader.

### Using an AVR ISP (In System Programmer)

- Your Arduino UNO (to program)
- An AVR Programmer such as the AVR Pocket Programmer
- An AVR Programming Cable (the pocket programmer comes with one)

If you have extra Arduino boards, but no ISP programmer, SparkFun.com has a cool tutorial on how to flash a bootloader using an Arduino as an ISP.

### Using another Arduino as an ISP

- Your Arduino UNO (to program)
- A Working Arduino (doesn't matter what kind)
- Some Male-to-Male Jumper Cables

For instructions on this method, take a look at the SparkFun.com website:  
<http://www.sparkfun.com/tutorials/247>

### Software Needed

ROBOTC is not currently able to burn a bootloader onto an Arduino board, so you'll need to download a copy of the latest version of the Arduino Open-Source programming language.

- Arduino Official Programming Language - Download Page

In addition, you'll need the ROBOTC modified bootloader. You can download that here:

- ROBOTC Modified UNO Bootloader - Modified Bootloader

### Bootload Download Instructions

- Download the Arduino Open Source Software and a copy of the Modified Bootloader File
- Copy the Modified Bootloader File into the /Arduino-1.0/hardware/arduino/bootloaders/stk500v2/ and overwrite the existing bootloader.

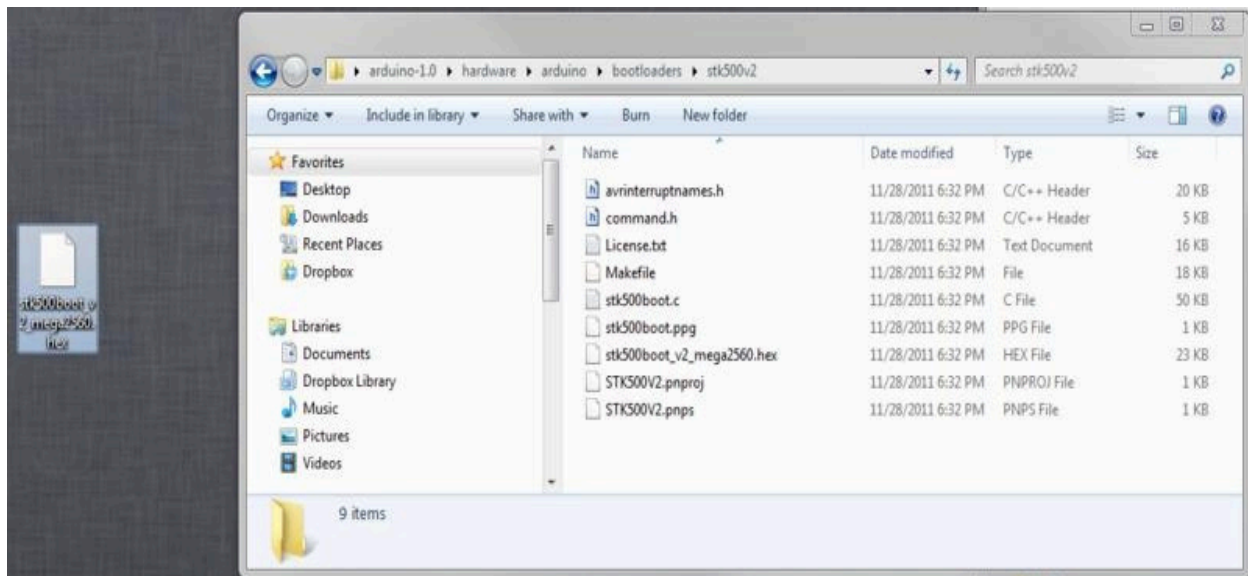


Fig 12 :Boot Download Instructor

- Power up your Arduino UNO (either via USB or external power)
- Plug in your AVR ISP Programmer to your computer (make sure you have any required drivers installed)
- Connect your AVR ISP Programmer into your Arduino UNO Board via the ISP Header (the 2x3 header pins right above the Arduino Logo)
- Launch the Arduino Open Source Software
- Change your settings in the Arduino Software to look for an Arduino UNO
- Change your settings in the Arduino Software to select your ISP Programmer Type (Check your programmer's documentation for the exact model)

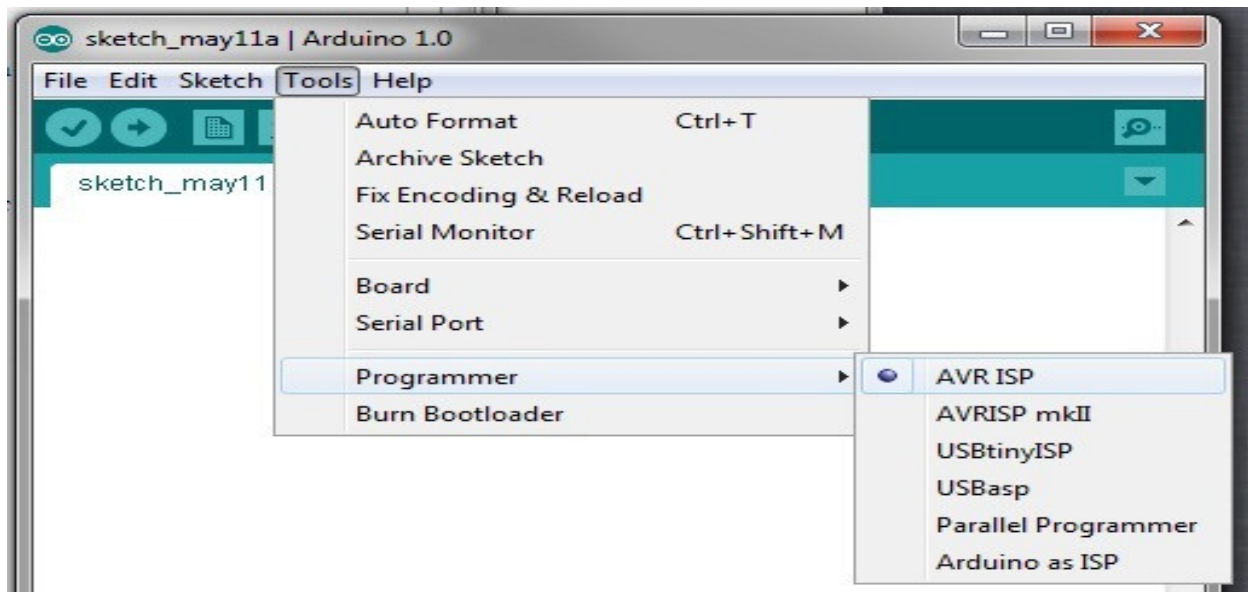


Fig. 13(a) select Tools menu

- Select the "Burn Bootloader" option under the "Tools" menu. The modified bootloader will now be sent to your Arduino. This typically take a minute or so.

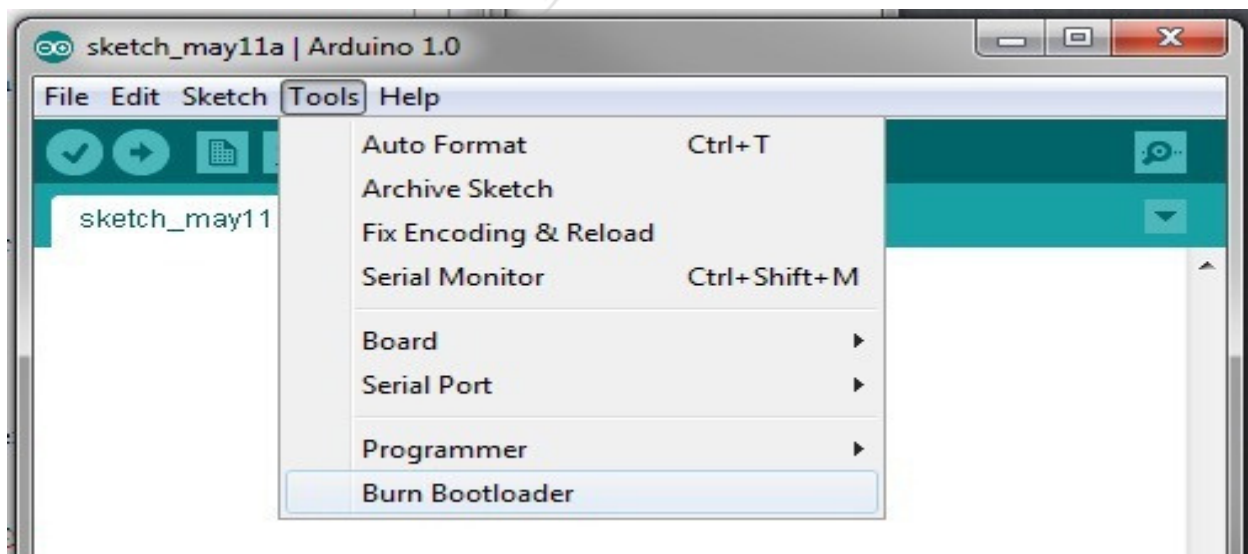


Fig. 13(b)select "Burn Bootloader" option under the "Tools" menu

- You should be all set to download ROBOTC firmware and start using your Arduino UNO with ROBOTC.

## Technical Details

The Arduino Boot loader sets the "erase Address" to zero every time the boot loader is called. ROBOTC called the "Load Address" command to set the address in which we want to write/verify when downloading program.

When writing a page of memory to the arduino, the Arduino boot loader will erase the existing page and write a whole new page.

In the scenario of downloading firmware, everything is great because the Erase Address and the Loaded Address both start at zero.

In the scenario of writing a user program, we start writing at memory location 0x7000, but the Bootloader erases information starting at location zero because the "Load Address" command doesn't update where to erase.

Our modification is to set both the Load Address and the Erase Address so the activity of writing a user program doesn't cause the firmware to be accidentally erased.

## Summary

|                        |                                     |
|------------------------|-------------------------------------|
| Microcontroller        | Arduino UNO                         |
| Operating Voltage      | 5V                                  |
| Input Voltage (limits) | 6-20V                               |
| Digital I/O Pins       | 54 (of which 14 provide PWM output) |

|                        |                                         |
|------------------------|-----------------------------------------|
| Analog Input Pins DC   | 16                                      |
| Current per I/O Pin DC | 40mA                                    |
| Current for 3.3V Pin   | 50mA                                    |
| Flash Memory SRAM      | 256 KB of which 8 KB used by bootloader |
| EEPROM Lock Speed      | 8KB                                     |
|                        | 4KB                                     |
|                        | 16MHz                                   |

The Arduino UNO can be powered via the USB connection or with an external power supply. The power source is selected automatically. External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

They differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the programmed as a USB-to-serial converter.

The power pins are as follows:

- $V_{IN}$  The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if

- supplying voltage via the power jack, access it through this pin.
- 5V. The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via a non-board regulator, or be supplied by USB or another regulated 5V supply.
- 3V3. A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50mA.
- GND. Ground pins.
- 

The ATMEGA has 256 KB of flash memory for storing code (of which 8 KB is used for the bootloader), 8 KB of SRAM and 4 KB of EEPROM (which can be read and written with the EEPROM library).

Each of the 54 digital pins on the Mega can be used as an input or output, using `pinMode()`, `digitalWrite()`, and `digitalRead()` functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50k Ohms. In addition, some pins have specialized functions:

- Serial: 0 (RX) and 1 (TX); Serial 1: 19 (RX) and 18 (TX); Serial 2: 17 (RX) and 16 (TX); Serial 3: 15 (RX) and 14 (TX).  
Used to receive (RX) and transmit (TX) TTL serial data. Pins 0 and 1 are also connected to the corresponding pins of the ATMEGA USB-to-TTL Serial chip.
- These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a changing value. See the `attachInterrupt()` function for details.

- PWM: 0 to 13. Provide 8-bit PWM output with the `analogWrite()` function.
- SPI: 50 (MISO), 51 (MOSI), 52 (SCK), 53 (SS). These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language. The SPI pins are also broken out on the ICSP header, which is physically compatible with the Duemilanove and Diecimila.
- LED: 13. There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.
- I2C: 2 (SDA), 3 (SCL) (cSODmAm) uanidca 2ti lo (nS CusLi)n. gS tuhpep ort I2 Wire library (documentation on the Wiring website). Note that these pins are not in the same location as the I2C pins on the Duemilanove.

The Arduino UNO has 16 analog inputs, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though it is possible to change the upper end of their range using the AREF pin and `analogReference()` function.

There are a couple of other pins on the board:

- AREF. Reference voltage for the analog inputs. Used with `analogReference()`.
- reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

## Communication

The Arduino UNO has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The Arduino UNO provides four hardware UARTs for TTL (5V) serial communication.

An ATMEGA on the board channels one of these over USB and provides a virtual comport to software on the computer (Windows machines will need a .inf file, but OSX and Linux machines will recognize the board as a COM port automatically. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the board. The RX and TX LEDs on the board will flash when data is being transmitted via the ATmega8U2 chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A SoftwareSerial library allows for serial communication on any of the digital pins.

The Arduino UNO also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation on the Wiring website for details. To use the SPI communication, please see the Arduino UNO datasheet.

## Programming

The Arduino UNO can be programmed with the Arduino software (download). For details, see the reference and tutorials.

The Arduino UNO comes pre-burned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files).

You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header; see these instructions for details.

## Automatic (Software) Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino UNO is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2 is connected to the reset line of the Arduino UNO via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino software uses this capability to allow you to upload code by simply pressing the upload button in the Arduino environment. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.

This setup has other implications. When the Arduino UNO is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half-second or so, the bootloader is running on the UNO. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts, make sure that the software with which it communicates waits a second after opening the connection and before sending this data.

The Mega contains a trace that can be cut to disable the auto-reset. The pads on either side of the trace can be soldered together to re-enable it. It's labeled "RESET-EN". You may also be able to disable the auto-reset by connecting a 110 ohm resistor from 5V to the reset line; see this forum thread for details.

## USB Over current Protection Physical Characteristics and Shield Compatibility

The Arduino UNO has a resettable poly fuse that protects your computer's USB ports from shorts and over current. Although most computers provide their own internal protection, the fuse provides an extra layer of protection. If more than 500mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed.

The maximum length and width of the UNO PCB are 4 and 2.1 inches respectively, with the USB connector and power jack extending beyond the former dimension. Three screw holes allow the board to be attached to a surface or case. Note that the distance between digital pins 7 and 8 is 160 mil (0.16"), not an even multiple of the 100 mil spacing of the other pins. The UNO is designed to be compatible with most shields designed for the Diecimila or Duemilanove. Digital pins 0 to 13 (and the adjacent AREF and GND pins), analog inputs 0 to 5, the power header, and ICSP header are all in equivalent locations. Further the main UART (serial port) is located on the same pins (0 and 1), as are external interrupts 0 and 1 (pins 2 and 3 respectively). SPI is available through the ICSP header on both the Mega and Duemilanove / Diecimila. Please note that I2C is not located on the same pins on the Mega (20 and 21) as the Duemilanove / Diecimila (analog inputs 4 and 5).

## How to use Arduino

Arduino can sense the environment by receiving input from a variety of sensors and can affect its surroundings by controlling lights, motors, and other actuators. The microcontroller on the board is programmed using the Arduino

programming language (based on Wiring) and the Arduino development environment (based on Processing). Arduino projects can be stand-alone or they can communicate with software running on a computer (e.g. Flash, Processing, MaxMSP).

Arduino is a cross-platform program. You'll have to follow different instructions for

your personal OS. Check on the Arduino site for the latest instructions.

*<http://arduino.cc/en/Guide/HomePage>*

Once you have downloaded/unzipped the arduino IDE, you can plug the Arduino to your PC via USB cable.

# PROGRAMMING CODE :

## TRANSMITTER SIDE :

```
#include <Keypad.h>
```

```
// Define keypad layout
```

```
const byte ROWS = 4;
```

```
const byte COLS = 4;
```

```
char keys[ROWS][COLS] = {
```

```
    {'1', '2', '3', 'A'},
```

```
    {'4', '5', '6', 'B'},
```

```
    {'7', '8', '9', 'C'},
```

```
    {'*', '0', '#', 'D'}
```

```
};
```

```
byte rowPins[ROWS] = {2, 3, 4, 5}; // Connect to row p
```

```
byte colPins[COLS] = {6, 7, 8, 9}; // Connect to column
```

```
pins
```

```
Keypad keypad = Keypad(makeKeymap(keys), rowPi
```

```
colPins, ROWS, COLS);
```

```
// LED pin
```

```
const int ledPin = 10;
```

# RECEIVER SIDE:

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
// Initialize LCD (address 0x27 is common; adjust if  
needed)
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
// LDR pin
```

```
const int ldrPin = A0;
```

```
void setup() {
```

```
  lcd.begin();
```

```
  lcd.backlight();
```

```
  lcd.setCursor(0, 0);
```

```
  lcd.print("Li-Fi Receiver");
```

```
  delay(2000);
```

```
  lcd.clear();
```

```
  pinMode(ldrPin, INPUT);
```

```
  Serial.begin(9600); // For debugging
```

```
}
```

```
void setup() {  
  pinMode(ledPin, OUTPUT);  
  digitalWrite(ledPin, LOW);  
  Serial.begin(9600); // For debugging  
}
```

```
void loop() {  
  char key = keypad.getKey();  
  
  if (key) {  
    Serial.println(key); // Debug  
    sendData(key);  
  }  
}
```

```
void sendData(char key) {  
  // Map each key to a unique pulse duration (in  
    microseconds)  
  unsign...
```

```
pinMode(ldrPin, INPUT);  
Serial.begin(9600); // For debugging  
}
```

```
void loop() {  
    // Read LDR value  
    int ldrValue = analogRead(ldrPin);  
  
    // Detect light pulse (threshold depends on your  
    // LDR and lighting)  
    if (ldrValue > 600) { // Adjust threshold based on  
        testing  
        unsigned long startTime = micros();  
        while (analogRead(ldrPin) > 600 && micros() -  
            startTime < 200000); // Wait for pulse end  
        unsigned long duration = micros() - startTime;  
  
        // Decode pulse duration to character  
        char receivedChar = decodePulse(duration);  
        if (receivedChar != 0) {  
            lcd.setCursor(0, 0);  
            lcd.print("Received: ");  
            lcd.print(receivedChar);  
            Serial.println(receivedChar); // Debug
```

```
}  
}  
}
```

```
char decodePulse(unsigned long duration) {  
    // Map pulse duration to character (match  
        transmitter durations)
```

```
    if (duration > 5000 && duration < 15000) return '1';  
    if (duration > 15000 && duration < 25000) return '2';  
    if (duration > 25000 && duration < 35000) return  
        '3';  
    if (duration > 35000 && duration < 45000) return  
        '4';  
    if (duration > 45000 && duration < 55000) return  
        '5';  
    if (duration > 55000 && duration < 65000) return  
        '6';  
    if (duration > 65000 && duration < 75000) return  
        '7';  
    if (duration > 75000 && duration < 85000) return  
        '8';  
    if (duration > 85000 && duration < 95000) return  
        '9';
```

```
        '9';
    if (duration > 95000 && duration < 105000) return
        '0';
    if (duration > 105000 && duration < 115000) return
        'A';
    if (duration > 115000 && duration < 125000) return
        'B';
    if (duration > 125000 && duration < 135000) return
        'C';
    if (duration > 135000 && duration < 145000) return
        'D';
    if (duration > 145000 && duration < 155000) return
        '*';
    if (duration > 155000 && duration < 165000) return
        '#';
    return 0; // Invalid duration
}
```

**THANKING  
YOU**